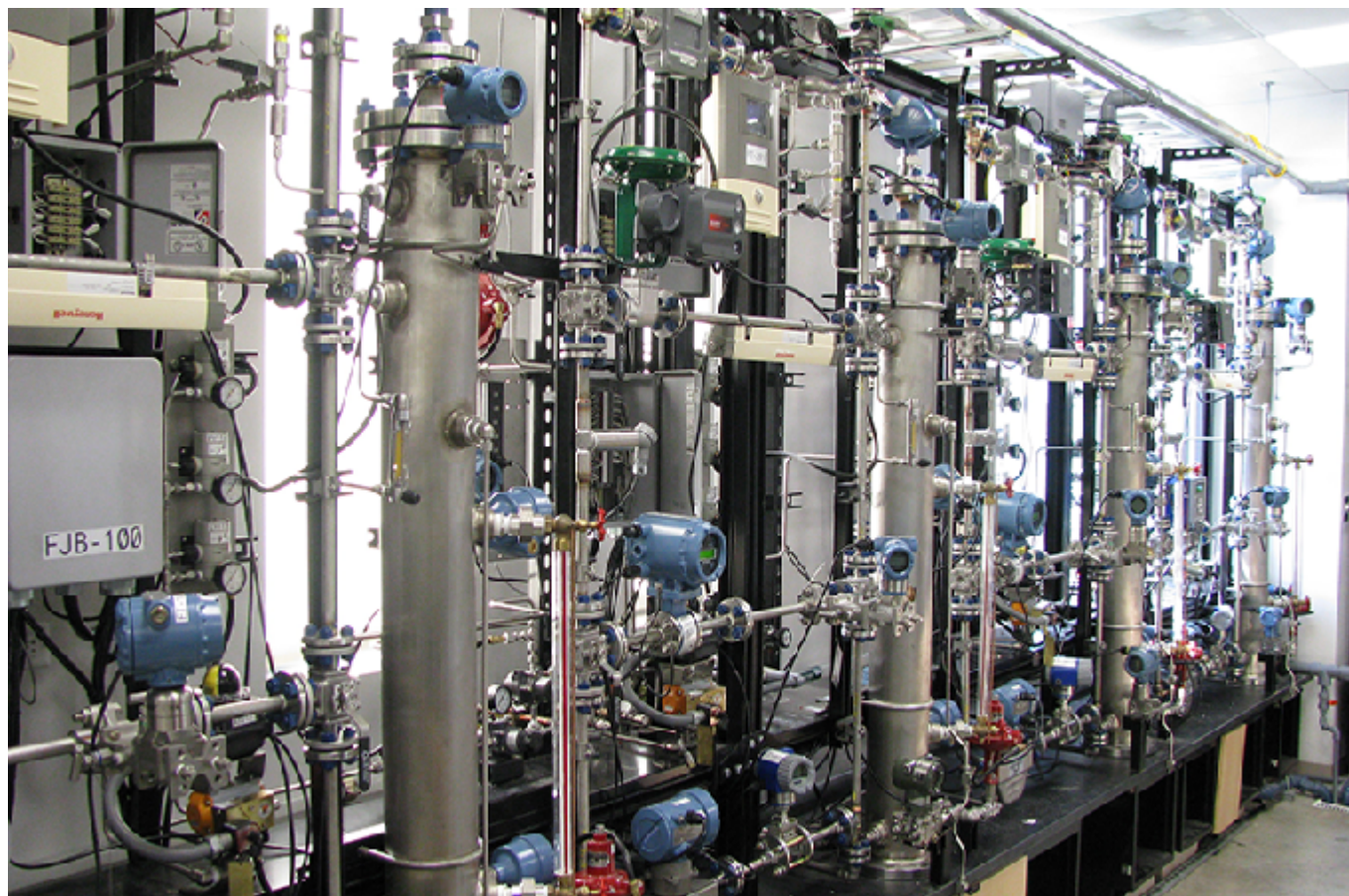




## Qualification Pack



# Assistant Instrumentation Mechanic (Metal Sector)

QP Code: ISC/Q1102

Version: 5.0

NSQF Level: 3

Indian Iron and Steel Sector Skill Council || Plot no.B-7, Action Area-III, 5th Floor, Room no.509, Near Hazra Kalibari, Opposite Uni World City, New Town, Kolkata, West Bengal  
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## Qualification Pack

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## Qualification Pack

### ISC/Q1102: Assistant Instrumentation Mechanic (Metal Sector)

#### Brief Job Description

The individual in this role performs installation, maintenance, testing and calibration of measuring and control equipment.

#### Personal Attributes

The person should be patient, organized, team-oriented and have the ability to work for long hours in adverse conditions. They must be able to plan and prioritize tasks effectively and have an eye for detail and quality.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [ISC/N0008: Use basic health and safety practices at the workplace](#)
2. [ISC/N0009: Work effectively with others](#)
3. [ISC/N1102: Carry out maintenance of measuring and control equipment](#)
4. [ISC/N1103: Perform inspection and calibration of measuring equipment](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

#### Qualification Pack (QP) Parameters

|                                      |                                                          |
|--------------------------------------|----------------------------------------------------------|
| <b>Sector</b>                        | Iron and Steel                                           |
| <b>Sub-Sector</b>                    | Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory |
| <b>Occupation</b>                    | Electronics and Instrumentation Maintenance              |
| <b>Country</b>                       | India                                                    |
| <b>NSQF Level</b>                    | 3                                                        |
| <b>Credits</b>                       | 14                                                       |
| <b>Aligned to NCO/ISCO/ISIC Code</b> | NCO-2015/7311.0401                                       |



## Qualification Pack

|                                                           |                                                                                                                                                                                                                            |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Minimum Educational Qualification &amp; Experience</b> | 10th Class<br>OR<br>9th Class with 1.5 years of experience in relevant field<br>OR<br>Grade 8 pass with 2 years of (NTC/ NAC) after 8th<br>OR<br>8th Class with 3 Years of experience in relevant field<br>OR<br>5th Class |
| <b>Minimum Level of Education for Training in School</b>  | Not Applicable                                                                                                                                                                                                             |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                         |
| <b>Minimum Job Entry Age</b>                              | 18 Years                                                                                                                                                                                                                   |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                         |
| <b>Next Review Date</b>                                   | 01/04/2028                                                                                                                                                                                                                 |
| <b>NSQC Approval Date</b>                                 | 01/04/2025                                                                                                                                                                                                                 |
| <b>Version</b>                                            | 5.0                                                                                                                                                                                                                        |
| <b>Reference code on NQR</b>                              | QG-03-IS-03881-2025-V2-IISSSC                                                                                                                                                                                              |
| <b>NQR Version</b>                                        | 2                                                                                                                                                                                                                          |



## Qualification Pack

### ISC/N0008: Use basic health and safety practices at the workplace

#### Description

This OS unit is about following safety and adopting sustainable practices for optimising use of resources.

#### Scope

The scope covers the following :

- Maintain safe and secure working environment
- Emergencies, rescue and first aid procedures
- Health and hygiene
- Housekeeping and waste management
- Material and energy conservation

#### Elements and Performance Criteria

##### *Maintain safe and secure working environment*

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous activities and the possible causes of risks or accidents in the workplace
- PC2.** follow safe working practices while dealing with hazards to ensure safety of self and others
- PC3.** use appropriate protective clothing/equipment for specific tasks and work
- PC4.** follow appropriate safety practices while working in and around trenches, elevated places and confined areas
- PC5.** lift heavy objects safely using correct procedures
- PC6.** carry out routine check of the machine for identifying potential hazards
- PC7.** report any identified breaches in health, safety and security policies and procedures to the designated person

##### *Emergencies, rescue and first aid procedures*

To be competent, the user/individual on the job must be able to:

- PC8.** use appropriate type of fire extinguisher
- PC9.** apply appropriate rescue techniques during fire hazard
- PC10.** provide appropriate first aid procedure to victims wherever required eg.in case of bleeding, burns, choking, electric shock etc.
- PC11.** follow emergency procedures such as raising alarm, safe evacuation etc.
- PC12.** attend safety training and fire drills to respond promptly during an emergency

##### *Health and hygiene*

To be competent, the user/individual on the job must be able to:

- PC13.** follow regular cleaning and disinfection practices at work place using appropriate techniques and materials
- PC14.** follow hand hygiene practices at work place using appropriate techniques and materials
- PC15.** report regarding the contagious illness of self or people in close contact
- PC16.** avoid contact with ill people and self-isolate in a similar situation





## Qualification Pack

### *Housekeeping and waste management*

To be competent, the user/individual on the job must be able to:

- PC17.** follow the fundamentals of 5S for housekeeping
- PC18.** ensure good housekeeping in order to prevent hazards and accidents
- PC19.** store the material, tools and equipment in the correct location and in good condition
- PC20.** segregate waste into different categories
- PC21.** identify recyclable, non-recyclable and hazardous waste
- PC22.** dispose non-recyclable, recyclable and reusable waste appropriately at identified location

### *Material and energy conservation*

To be competent, the user/individual on the job must be able to:

- PC23.** identify ways to optimize usage of material in various tasks/activities/processes
- PC24.** check for spills/leakages in various tasks/activities/processes
- PC25.** plug spills/leakages and escalate to appropriate authority if unable to rectify
- PC26.** check if the equipment/machine is functioning normally before commencing work and rectify wherever required
- PC27.** ensure electrical equipment and appliances are properly connected and turned off when not in use

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context
- KU2.** the organisation's emergency procedures for different emergency situations and the importance of following the same
- KU3.** evacuation procedures for workers and visitors
- KU4.** how and when to report hazards
- KU5.** potential hazards, risks and threats based on the nature of work
- KU6.** preventative and remedial actions to be taken in case of exposure to toxic material
- KU7.** various types of fire extinguisher
- KU8.** various types of safety signs and their meaning
- KU9.** appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.
- KU10.** relevant standards, procedures and policies related to 5S followed in the company
- KU11.** the various materials used and their storage norms
- KU12.** efficient utilisation of material and water
- KU13.** basics of electricity and prevalent energy efficient devices
- KU14.** common practices of conserving electricity
- KU15.** categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU16.** usage of different colors of dustbins



## Qualification Pack

**KU17.** waste management techniques

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** ask for clarifications from superior about the job requirement
- GS4.** work with supervisors/team members to carry out work related tasks
- GS5.** complete tasks efficiently and accurately within stipulated time
- GS6.** inform/report to concerned person in case of any problem
- GS7.** make timely decisions for efficient utilization of resources
- GS8.** write reports such as accident report, in at least English/regional language
- GS9.** be punctual and utilize time efficiently



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Maintain safe and secure working environment</i>                                                                                           | <b>10</b>    | <b>14</b>       | -             | <b>6</b>   |
| <b>PC1.</b> identify hazardous activities and the possible causes of risks or accidents in the workplace                                      | 2            | 2               | -             | 1          |
| <b>PC2.</b> follow safe working practices while dealing with hazards to ensure safety of self and others                                      | 2            | 3               | -             | 1          |
| <b>PC3.</b> use appropriate protective clothing/equipment for specific tasks and work                                                         | 1            | 2               | -             | 1          |
| <b>PC4.</b> follow appropriate safety practices while working in and around trenches, elevated places and confined areas                      | 2            | 1               | -             | -          |
| <b>PC5.</b> lift heavy objects safely using correct procedures                                                                                | 1            | 2               | -             | 1          |
| <b>PC6.</b> carry out routine check of the machine for identifying potential hazards                                                          | 1            | 2               | -             | 1          |
| <b>PC7.</b> report any identified breaches in health, safety and security policies and procedures to the designated person                    | 1            | 2               | -             | 1          |
| <i>Emergencies, rescue and first aid procedures</i>                                                                                           | <b>6</b>     | <b>9</b>        | -             | <b>5</b>   |
| <b>PC8.</b> use appropriate type of fire extinguisher                                                                                         | 1            | 1               | -             | 1          |
| <b>PC9.</b> apply appropriate rescue techniques during fire hazard                                                                            | 1            | 2               | -             | 1          |
| <b>PC10.</b> provide appropriate first aid procedure to victims wherever required eg.in case of bleeding, burns, choking, electric shock etc. | 2            | 2               | -             | 1          |
| <b>PC11.</b> follow emergency procedures such as raising alarm, safe evacuation etc.                                                          | 1            | 2               | -             | 1          |
| <b>PC12.</b> attend safety training and fire drills to respond promptly during an emergency                                                   | 1            | 2               | -             | 1          |
| <i>Health and hygiene</i>                                                                                                                     | <b>2</b>     | <b>6</b>        | -             | <b>2</b>   |





## Qualification Pack

| Assessment Criteria for Outcomes                                                                                         | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> follow regular cleaning and disinfection practices at work place using appropriate techniques and materials | 1            | 2               | -             | 1          |
| <b>PC14.</b> follow hand hygiene practices at work place using appropriate techniques and materials                      | 1            | 2               | -             | 1          |
| <b>PC15.</b> report regarding the contagious illness of self or people in close contact                                  | -            | 1               | -             | -          |
| <b>PC16.</b> avoid contact with ill people and self-isolate in a similar situation                                       | -            | 1               | -             | -          |
| <i>Housekeeping and waste management</i>                                                                                 | <b>7</b>     | <b>12</b>       | -             | <b>5</b>   |
| <b>PC17.</b> follow the fundamentals of 5S for housekeeping                                                              | 2            | 3               | -             | 2          |
| <b>PC18.</b> ensure good housekeeping in order to prevent hazards and accidents                                          | 1            | 2               | -             | -          |
| <b>PC19.</b> store the material, tools and equipment in the correct location and in good condition                       | 1            | 2               | -             | -          |
| <b>PC20.</b> segregate waste into different categories                                                                   | 1            | 2               | -             | 1          |
| <b>PC21.</b> identify recyclable, non-recyclable and hazardous waste                                                     | 1            | 1               | -             | 1          |
| <b>PC22.</b> dispose non-recyclable, recyclable and reusable waste appropriately at identified location                  | 1            | 2               | -             | 1          |
| <i>Material and energy conservation</i>                                                                                  | <b>5</b>     | <b>9</b>        | -             | <b>2</b>   |
| <b>PC23.</b> identify ways to optimize usage of material in various tasks/activities/processes                           | 1            | 2               | -             | -          |
| <b>PC24.</b> check for spills/leakages in various tasks/activities/processes                                             | 1            | 2               | -             | 1          |
| <b>PC25.</b> plug spills/leakages and escalate to appropriate authority if unable to rectify                             | 1            | 2               | -             | 1          |
| <b>PC26.</b> check if the equipment/machine is functioning normally before commencing work and rectify wherever required | 1            | 2               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC27.</b> ensure electrical equipment and appliances are properly connected and turned off when not in use | 1            | 1               | -             | -          |
| <b>NOS Total</b>                                                                                              | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                                                                                                                                          |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | ISC/N0008                                                                                                                                                                                                |
| <b>NOS Name</b>            | Use basic health and safety practices at the workplace                                                                                                                                                   |
| <b>Sector</b>              | Iron and Steel                                                                                                                                                                                           |
| <b>Sub-Sector</b>          | Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory                                                                                                                                                 |
| <b>Occupation</b>          | Mechanical Maintenance, Raw Material Handling , Product Quality Control, Refractory Maintenance, Furnace Operation, Fitter, Cold Rolling Mill Operations, Electrical Maintenance, Mechanical Maintenance |
| <b>NSQF Level</b>          | 3                                                                                                                                                                                                        |
| <b>Credits</b>             | TBD                                                                                                                                                                                                      |
| <b>Version</b>             | 2.0                                                                                                                                                                                                      |
| <b>Last Reviewed Date</b>  | 12/03/2026                                                                                                                                                                                               |
| <b>Next Review Date</b>    | 12/09/2026                                                                                                                                                                                               |
| <b>NSQC Clearance Date</b> | 12/03/2026                                                                                                                                                                                               |



## Qualification Pack

### ISC/N0009: Work effectively with others

#### Description

This OS unit is about communicating with colleagues/superiors and others, either in own work group or in other work groups within organisation.

#### Scope

The scope covers the following :

- Communicate effectively with colleagues and others
- Interact with supervisor
- Follow appropriate behaviour at work place

#### Elements and Performance Criteria

##### *Communicate effectively with colleagues and others*

To be competent, the user/individual on the job must be able to:

- PC1.** coordinate with colleagues to share work, as per the workload in order to achieve team goals
- PC2.** maintain clear communication with colleagues and others, wherever needed, through all means i.e. face-to-face, telephonic or written
- PC3.** adjust communication styles to reflect gender and persons with disability (PwD) sensitivity
- PC4.** respect all colleagues and co-workers
- PC5.** resolve conflicts by communicating with colleagues and other departments

##### *Interact with supervisor*

To be competent, the user/individual on the job must be able to:

- PC6.** identify work requirements by receiving instructions from reporting supervisor
- PC7.** escalate problems to supervisors that cannot be handled
- PC8.** report the completed work
- PC9.** interact with the reporting supervisor about any possible hazards and safety concerns

##### *Follow appropriate behaviour at work place*

To be competent, the user/individual on the job must be able to:

- PC10.** extend help to people with Disability (PwD) at workplace, if required
- PC11.** empathize with people with disability
- PC12.** adopt a gender neutral behavior
- PC13.** adopt responsible and disciplined behaviours at the workplace

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of effective communication and establishing good working relationships with colleagues and supervisor



## Qualification Pack

- KU2.** different methods of communication as per the circumstances
- KU3.** importance of teamwork in organization and individual success
- KU4.** various components of effective communication
- KU5.** barriers to effective communication
- KU6.** common reasons for interpersonal conflict
- KU7.** what constitutes disciplined behaviour for a working professional
- KU8.** gender concepts, issues & legislation
- KU9.** organisational policies and procedures related to gender equality
- KU10.** challenges faced by PWD and the ways to help them overcome the same

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read instructions/guidelines/procedures
- GS2.** listen effectively and orally communicate information
- GS3.** ask for clarification and advice from the concerned person
- GS4.** maintain positive and effective relationships with colleagues
- GS5.** evaluate the possible solution(s) to the problem
- GS6.** spot and communicate potential areas of disruptions in the work process and report the same
- GS7.** complete written work with attention to detail
- GS8.** check that the work meets customer requirements



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Communicate effectively with colleagues and others</i>                                                                                        | <b>13</b>    | <b>20</b>       | -             | <b>9</b>   |
| <b>PC1.</b> coordinate with colleagues to share work, as per the workload in order to achieve team goals                                         | 3            | 5               | -             | 2          |
| <b>PC2.</b> maintain clear communication with colleagues and others, wherever needed, through all means i.e. face-to-face, telephonic or written | 5            | 7               | -             | 3          |
| <b>PC3.</b> adjust communication styles to reflect gender and persons with disability (PwD) sensitivity                                          | 3            | 4               | -             | 2          |
| <b>PC4.</b> respect all colleagues and co-workers                                                                                                | 1            | 2               | -             | 1          |
| <b>PC5.</b> resolve conflicts by communicating with colleagues and other departments                                                             | 1            | 2               | -             | 1          |
| <i>Interact with supervisor</i>                                                                                                                  | <b>8</b>     | <b>14</b>       | -             | <b>6</b>   |
| <b>PC6.</b> identify work requirements by receiving instructions from reporting supervisor                                                       | 2            | 3               | -             | 1          |
| <b>PC7.</b> escalate problems to supervisors that cannot be handled                                                                              | 2            | 3               | -             | 2          |
| <b>PC8.</b> report the completed work                                                                                                            | 2            | 3               | -             | 1          |
| <b>PC9.</b> interact with the reporting supervisor about any possible hazards and safety concerns                                                | 2            | 5               | -             | 2          |
| <i>Follow appropriate behaviour at work place</i>                                                                                                | <b>9</b>     | <b>16</b>       | -             | <b>5</b>   |
| <b>PC10.</b> extend help to people with Disability (PwD) at workplace, if required                                                               | 2            | 4               | -             | 2          |
| <b>PC11.</b> empathize with people with disability                                                                                               | 2            | 4               | -             | 1          |
| <b>PC12.</b> adopt a gender neutral behavior                                                                                                     | 2            | 4               | -             | 1          |
| <b>PC13.</b> adopt responsible and disciplined behaviours at the workplace                                                                       | 3            | 4               | -             | 1          |





## Qualification Pack

| Assessment Criteria for Outcomes | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------|--------------|-----------------|---------------|------------|
| NOS Total                        | 30           | 50              | -             | 20         |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | ISC/N0009                                                                                                                                                                   |
| <b>NOS Name</b>            | Work effectively with others                                                                                                                                                |
| <b>Sector</b>              | Iron and Steel                                                                                                                                                              |
| <b>Sub-Sector</b>          | Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory                                                                                                                    |
| <b>Occupation</b>          | Mechanical Maintenance, Raw Material Handling , Product Quality Control, Refractory Maintenance, Furnace Operation, Fitter, Operations, Coke Making, Mechanical Maintenance |
| <b>NSQF Level</b>          | 3                                                                                                                                                                           |
| <b>Credits</b>             | TBD                                                                                                                                                                         |
| <b>Version</b>             | 2.0                                                                                                                                                                         |
| <b>Last Reviewed Date</b>  | 12/03/2026                                                                                                                                                                  |
| <b>Next Review Date</b>    | 12/09/2026                                                                                                                                                                  |
| <b>NSQC Clearance Date</b> | 12/03/2026                                                                                                                                                                  |



## Qualification Pack

# ISC/N1102: Carry out maintenance of measuring and control equipment

## Description

This OS unit is about assisting the "Technician Instrumentation" during maintenance of measuring and control equipment, in accordance with approved procedures.

## Scope

The scope covers the following :

- Prepare for maintenance work
- Perform maintenance of measuring and control equipment
- Perform post-maintenance activities

## Elements and Performance Criteria

### *Prepare for maintenance work*

To be competent, the user/individual on the job must be able to:

- PC1.** identify the measuring and control equipment, its manufacturer's specifications and functioning from the user manual
- PC2.** read maintenance schedule and checklist and plan the time and schedule for conducting the maintenance
- PC3.** identify and arrange the tools, measuring instruments, equipment and spares required for the job

### *Perform maintenance of measuring and control equipment*

To be competent, the user/individual on the job must be able to:

- PC4.** check from "Technician Instrumentation" that the measuring device is functioning within tolerance limits or not
- PC5.** re-fix or re-position the measuring device until reading comes to the satisfying range, in case of errors in reading
- PC6.** connect the linking device so that no transmission errors take place due to intermediate losses / interference
- PC7.** perform basic health check-up of measuring and control equipment as specified in the maintenance checklist
- PC8.** execute suitable re-routing of transmission system if need to achieve satisfactory results
- PC9.** check the equipment parts for any wear and tear, damage, improper functioning etc.
- PC10.** repair or replace damaged components as per the requirement
- PC11.** re-connect or return the system after completion of maintenance activities
- PC12.** report any instances where the maintenance activities cannot be fully met or where there are identified defects outside the planned schedule
- PC13.** record all repairs carried out, time taken and unplanned tasks encountered during the maintenance activities
- PC14.** dispose off waste materials such as failed parts/aggregates, as per organisation's policies

### *Perform post-maintenance activities*



## Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC15.** re-position the pick-up / sensor to better location as advised by “Technician Instrumentation”
- PC16.** re-fix the pick-up / sensor with better fixing device / fastener as advised by “Technician Instrumentation”
- PC17.** monitor the problem and keep the superior informed about progress or any delays in resolving the problem
- PC18.** refer the problem to “Technician Instrumentation” or competent internal / external specialist if it cannot be resolved

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** Standard Operating Procedures (SOP) recommended by the manufacturer for using hand tools, measuring instruments and equipment required during maintenance work
- KU3.** isolation and lock-off procedures or permit-to-work procedure that applies
- KU4.** functioning of different process plant and its measuring and control equipment
- KU5.** various equipment i.e. Pressure (e.g. absolute, gauge, vacuum), Flow (e.g. orifice plate, venturi tube, electromagnetic, ultrasonic, differential pressure cell, positive displacement), Level (e.g. floats, displacer, differential pressure cells, load cells, ultrasonic, conductivity), Temperature (e.g. bi-metallic, thermocouples, resistance, infra-red, thermal imaging), Weight (e.g. mechanical systems, load cells/strain gauges, transducers), Fiscal metering (e.g. gas, electricity, water, fuel), Detection and alarm (e.g. smoke, heat, gas, chemical, water, metal), Speed measurement (e.g. mechanical, electrical, stroboscopic), Emergency shutdown, Speed control (e.g. mechanical governors, electrical governors, DC speed controller, AC motor control systems, stepper motors, invertors), Vibration monitoring (e.g. vibration switches, proximity probes, seismic velocity transducer, linear variable differential transformers, portable data collectors), Analyzers (e.g. gas detection, spectroscopy, oxygen analyzer, water analysis, moisture measurement, density), Recorders and indicators, Telemetry systems (e.g. master station, outstation, standalone systems), Valves and valve mechanisms (e.g. control valves, valve actuators and positioners), Other specific instrumentation equipment
- KU6.** how to analyze evidence and evaluate possible characteristics and causes of specific faults/problems
- KU7.** how to relate previous reports/records of similar fault conditions
- KU8.** how to handle specific sensitive devices / sensors safely
- KU9.** precautions to be taken to prevent electrostatic discharge (ESD) damage to electronic circuits and components
- KU10.** basic principles of operation of the instrumentation and control equipment being maintained
- KU11.** correct way of fitting sensors to avoid faulty readings (caused by head correction, poor flow past sensor, blockages, incorrect wiring, poor insulation or incorrect materials)
- KU12.** correct and tidy installation and connection of external wiring and components, to avoid electronic interference or mechanical damage
- KU13.** how to carry out visual checks of the instruments (e.g. checking for leaks, security of joints and physical damage)



## Qualification Pack

**KU14.** safety requirements to be followed during the maintenance work

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognise a workplace problem and take suitable action
- GS6.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organise work according to the work requirements
- GS8.** complete the assigned tasks with minimum supervision
- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Prepare for maintenance work</i>                                                                                              | <b>5</b>     | <b>6</b>        | -             | <b>4</b>   |
| <b>PC1.</b> identify the measuring and control equipment, its manufacturer's specifications and functioning from the user manual | 1            | 2               | -             | 1          |
| <b>PC2.</b> read maintenance schedule and checklist and plan the time and schedule for conducting the maintenance                | 1            | 2               | -             | 1          |
| <b>PC3.</b> identify and arrange the tools, measuring instruments, equipment and spares required for the job                     | 3            | 2               | -             | 2          |
| <i>Perform maintenance of measuring and control equipment</i>                                                                    | <b>17</b>    | <b>32</b>       | -             | <b>11</b>  |
| <b>PC4.</b> check from "Technician Instrumentation" that the measuring device is functioning within tolerance limits or not      | 1            | 2               | -             | 1          |
| <b>PC5.</b> re-fix or re-position the measuring device until reading comes to the satisfying range, in case of errors in reading | 2            | 3               | -             | 1          |
| <b>PC6.</b> connect the linking device so that no transmission errors take place due to intermediate losses / interference       | 1            | 2               | -             | 1          |
| <b>PC7.</b> perform basic health check-up of measuring and control equipment as specified in the maintenance checklist           | 3            | 5               | -             | 2          |
| <b>PC8.</b> execute suitable re-routing of transmission system if need to achieve satisfactory results                           | 1            | 2               | -             | 1          |
| <b>PC9.</b> check the equipment parts for any wear and tear, damage, improper functioning etc.                                   | 2            | 5               | -             | 1          |
| <b>PC10.</b> repair or replace damaged components as per the requirement                                                         | 3            | 5               | -             | 2          |
| <b>PC11.</b> re-connect or return the system after completion of maintenance activities                                          | 1            | 2               | -             | 1          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> report any instances where the maintenance activities cannot be fully met or where there are identified defects outside the planned schedule | 1            | 2               | -             | -          |
| <b>PC13.</b> record all repairs carried out, time taken and unplanned tasks encountered during the maintenance activities                                 | 1            | 2               | -             | 1          |
| <b>PC14.</b> dispose off waste materials such as failed parts/aggregates, as per organisation's policies                                                  | 1            | 2               | -             | -          |
| <i>Perform post-maintenance activities</i>                                                                                                                | <b>8</b>     | <b>12</b>       | -             | <b>5</b>   |
| <b>PC15.</b> re-position the pick-up / sensor to better location as advised by "Technician Instrumentation"                                               | 2            | 3               | -             | 1          |
| <b>PC16.</b> re-fix the pick-up / sensor with better fixing device / fastener as advised by "Technician Instrumentation"                                  | 2            | 3               | -             | 1          |
| <b>PC17.</b> monitor the problem and keep the superior informed about progress or any delays in resolving the problem                                     | 2            | 3               | -             | 1          |
| <b>PC18.</b> refer the problem to "Technician Instrumentation" or competent internal / external specialist if it cannot be resolved                       | 2            | 3               | -             | 2          |
| <b>NOS Total</b>                                                                                                                                          | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                       |
|----------------------------|-----------------------------------------------------------------------|
| <b>NOS Code</b>            | ISC/N1102                                                             |
| <b>NOS Name</b>            | Carry out maintenance of measuring and control equipment              |
| <b>Sector</b>              | Iron and Steel                                                        |
| <b>Sub-Sector</b>          | Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory, Iron Making |
| <b>Occupation</b>          | Electronics and Instrumentation Maintenance                           |
| <b>NSQF Level</b>          | 3                                                                     |
| <b>Credits</b>             | TBD                                                                   |
| <b>Version</b>             | 2.0                                                                   |
| <b>Last Reviewed Date</b>  | 01/04/2025                                                            |
| <b>Next Review Date</b>    | 01/04/2028                                                            |
| <b>NSQC Clearance Date</b> | 01/04/2025                                                            |





## Qualification Pack

# ISC/N1103: Perform inspection and calibration of measuring equipment

## Description

This OS unit is about testing and calibration of measuring and control equipment for correct operation in accordance with pre-determined procedures.

## Scope

The scope covers the following :

- Testing of measuring and control equipment
- Calibrating measuring and control equipment

## Elements and Performance Criteria

### *Testing of measuring and control equipment*

To be competent, the user/individual on the job must be able to:

- PC1.** identify the testing and calibration requirements by following standard operating procedures
- PC2.** prepare and update testing/calibration schedules and plans as per the WI
- PC3.** select the appropriate test equipment in accordance with defined requirements
- PC4.** select the correct test applications for testing of measuring and control equipment
- PC5.** check the components, leads, fasteners, etc. for any damage, wear and tear, loose connections and other faults
- PC6.** carry out testing of the measuring and control equipment i.e. sensors, transmitters, converters, indicators, analyzers, controllers, power supplies, circuit boards etc. to diagnose faults by using testing devices
- PC7.** apply appropriate testing procedures for assessing the operation of measuring and control equipment
- PC8.** record observations/ readings as per the parameters mentioned in the testing manual/Work Instructions
- PC9.** analyze the test results against specified operational specifications and localized faults and report to the supervisor about the potential faults identified
- PC10.** suggest and implement corrective actions to rectify the issues identified as per SOP
- PC11.** monitor and review the effectiveness of corrective actions planned for fault rectification and prepare reports for the regulatory authorities on the same

### *Calibrating measuring and control equipment*

To be competent, the user/individual on the job must be able to:

- PC12.** assess the calibration of measuring and control equipment w.r.t manufacturers' specifications and/or standard operating procedures
- PC13.** carry out calibration of each measuring and control equipment against appropriate physical standards by using correct calibration devices, equipment, techniques and predetermined procedures
- PC14.** check zero error, span and range of the equipment on indicators/controllers by using correct and appropriate configuration



## Qualification Pack

- PC15.** prepare and update documents and records related to testing/calibration activities done
- PC16.** carry out recommissioning of equipment in accordance with standard operating procedures
- PC17.** report any instances where the testing/calibration activities cannot be fully met

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** SOPs for checking and calibration of measuring equipment
- KU3.** Standard Operating Procedures (SOP) recommended by OEM for using tools and equipment to check measuring equipment for faults
- KU4.** correct operation of the instrumentation system including the procedures for isolating instrumentation systems
- KU5.** various techniques to check the calibration of the measuring equipment for conformance to specifications
- KU6.** how to inspect and test the instrumentation system
- KU7.** various types of tests i.e. Visual inspection of the instrument for completeness and freedom from damage or foreign objects, Standard serviceability test/calibration, Equipment self-diagnostics, Leak/pressure test, Signal injection tests, Soak test, Special-to-type tests, Signal measurement and transmission, Operational/function checks, Five point calibration, Unit substitution etc. need to conduct
- KU8.** various components i.e. Sensors, Transmitters, Converters, Indicators, Analyzers, Controllers, Power supplies, Removable circuit boards, Sensor units associated with determining/controlling density, level, flow, temperature, composition etc. of a range of materials need to be tested
- KU9.** various equipment i.e. Pressure (e.g. absolute, gauge, vacuum), Flow (e.g. orifice plate, venturi tube, electromagnetic, ultrasonic, differential pressure cell, positive displacement), Level (e.g. floats, displacer, differential pressure cells, load cells, ultrasonic, conductivity), Temperature (e.g. bi-metallic, thermocouples, resistance, infra-red, thermal imaging), Weight (e.g. mechanical systems, load cells/strain gauges, transducers), Fiscal metering (e.g. gas, electricity, water, fuel), Detection and alarm (e.g. smoke, heat, gas, chemical, water, metal), Speed measurement (e.g. mechanical, electrical, stroboscopic), Emergency shutdown, Speed control (e.g. mechanical governors, electrical governors, DC speed controller, AC motor control systems, stepper motors, invertors), Vibration monitoring (e.g. vibration switches, proximity probes, seismic velocity transducer, linear variable differential transformers, portable data collectors), Analyzers (e.g. gas detection, spectroscopy, oxygen analyzer, water analysis, moisture measurement, density), Recorders and indicators, Telemetry systems (e.g. master station, outstation, standalone systems), Valves and valve mechanisms (e.g. control valves, valve actuators and positioners), Other specific instrumentation equipment



## Qualification Pack

- KU10.** various calibration tools used are Oscilloscopes, Pressure gauge, Standard test gauges, Temperature controllers, Temperature baths, Micrometer, Current injection devices, Voltmeter, All types of comparators, Jigs and fixtures, Templates and patterns, Insulation testers, Calibrated weights, Pressure sources, Vernier calliper, Analogue and digital meters, Digital pressure indicators, Dead weight tester, Logic probes, Calibrated flow meters, Special purpose test equipment, System calibrators, Manometers, pH simulator/buffers, Wheatstone bridge, Potentiometers, Frequency/ signal generators, Logic probes, Multimeters, (analog/digital), Test gauges, Cathode ray oscilloscopes and other associated equipment
- KU11.** procedures for checking and verifying the operational function of the instrumentation system/equipment
- KU12.** specifications of each instrumentation system and acceptable deviations from specifications
- KU13.** probable causes of faults in instrumentation system/equipment components
- KU14.** procedures for repairing faulty instrumentation system
- KU15.** procedures for zero error, span and range checks on instrumentation systems/equipment
- KU16.** how to calibrate the measuring equipment against the appropriate physical standard
- KU17.** how to re-commission the measuring equipment
- KU18.** checks that are to be made on measuring equipment to identify faults
- KU19.** common fault(s) that may be found in the measuring equipment
- KU20.** effects of faults on the performance/accuracy of the measuring equipment
- KU21.** purpose/operational function of instrumentation system
- KU22.** functionality of the equipment and tolerance levels for calibration
- KU23.** various instrumentation principles (e.g. controlling density, level, flow, temperature, composition of a range of materials)
- KU24.** interpretation requirements of schematic, wiring and block diagrams and circuits
- KU25.** principles of hydraulic, pneumatic and electrical flow

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognise a workplace problem and take suitable action
- GS6.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organise work according to the work requirements
- GS8.** complete the assigned tasks with minimum supervision
- GS9.** complete the assigned tasks with minimum supervision



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Testing of measuring and control equipment</i>                                                                                                                                                                                | <b>20</b>    | <b>32</b>       | <b>-</b>      | <b>13</b>  |
| <b>PC1.</b> identify the testing and calibration requirements by following standard operating procedures                                                                                                                         | 1            | 2               | -             | -          |
| <b>PC2.</b> prepare and update testing/calibration schedules and plans as per the WI                                                                                                                                             | 1            | 2               | -             | 1          |
| <b>PC3.</b> select the appropriate test equipment in accordance with defined requirements                                                                                                                                        | 2            | 2               | -             | 2          |
| <b>PC4.</b> select the correct test applications for testing of measuring and control equipment                                                                                                                                  | 2            | 2               | -             | 1          |
| <b>PC5.</b> check the components, leads, fasteners, etc. for any damage, wear and tear, loose connections and other faults                                                                                                       | 2            | 3               | -             | 1          |
| <b>PC6.</b> carry out testing of the measuring and control equipment i.e. sensors, transmitters, converters, indicators, analyzers, controllers, power supplies, circuit boards etc. to diagnose faults by using testing devices | 3            | 5               | -             | 2          |
| <b>PC7.</b> apply appropriate testing procedures for assessing the operation of measuring and control equipment                                                                                                                  | 3            | 5               | -             | 2          |
| <b>PC8.</b> record observations/ readings as per the parameters mentioned in the testing manual/Work Instructions                                                                                                                | 1            | 2               | -             | 1          |
| <b>PC9.</b> analyze the test results against specified operational specifications and localized faults and report to the supervisor about the potential faults identified                                                        | 2            | 4               | -             | 1          |
| <b>PC10.</b> suggest and implement corrective actions to rectify the issues identified as per SOP                                                                                                                                | 2            | 3               | -             | 1          |
| <b>PC11.</b> monitor and review the effectiveness of corrective actions planned for fault rectification and prepare reports for the regulatory authorities on the same                                                           | 1            | 2               | -             | 1          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Calibrating measuring and control equipment</i>                                                                                                                                                         | <b>10</b>    | <b>18</b>       | -             | <b>7</b>   |
| <b>PC12.</b> assess the calibration of measuring and control equipment w.r.t manufacturers' specifications and/or standard operating procedures                                                            | 2            | 3               | -             | 1          |
| <b>PC13.</b> carry out calibration of each measuring and control equipment against appropriate physical standards by using correct calibration devices, equipment, techniques and predetermined procedures | 3            | 5               | -             | 2          |
| <b>PC14.</b> check zero error, span and range of the equipment on indicators/controllers by using correct and appropriate configuration                                                                    | 2            | 4               | -             | 2          |
| <b>PC15.</b> prepare and update documents and records related to testing/calibration activities done                                                                                                       | 1            | 2               | -             | 1          |
| <b>PC16.</b> carry out recommissioning of equipment in accordance with standard operating procedures                                                                                                       | 1            | 2               | -             | -          |
| <b>PC17.</b> report any instances where the testing/calibration activities cannot be fully met                                                                                                             | 1            | 2               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                                                                           | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                           |
|----------------------------|-----------------------------------------------------------|
| <b>NOS Code</b>            | ISC/N1103                                                 |
| <b>NOS Name</b>            | Perform inspection and calibration of measuring equipment |
| <b>Sector</b>              | Iron and Steel                                            |
| <b>Sub-Sector</b>          | Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory  |
| <b>Occupation</b>          | Electronics and Instrumentation Maintenance               |
| <b>NSQF Level</b>          | 3                                                         |
| <b>Credits</b>             | TBD                                                       |
| <b>Version</b>             | 2.0                                                       |
| <b>Last Reviewed Date</b>  | 01/04/2025                                                |
| <b>Next Review Date</b>    | 01/04/2028                                                |
| <b>NSQC Clearance Date</b> | 01/04/2025                                                |



## Qualification Pack

### DGT/VSQ/N0101: Employability Skills (30 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

**PC1.** understand the significance of employability skills in meeting the job requirements

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

**PC2.** identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

**PC3.** explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

**PC4.** speak with others using some basic English phrases or sentences

##### *Communication Skills*

To be competent, the user/individual on the job must be able to:

**PC5.** follow good manners while communicating with others

**PC6.** work with others in a team



## Qualification Pack

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

**PC7.** communicate and behave appropriately with all genders and PwD

**PC8.** report any issues related to sexual harassment

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

**PC9.** use various financial products and services safely and securely

**PC10.** calculate income, expenses, savings etc.

**PC11.** approach the concerned authorities for any exploitation as per legal rights and laws

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

**PC12.** operate digital devices and use its features and applications securely and safely

**PC13.** use internet and social media platforms securely and safely

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

**PC14.** identify and assess opportunities for potential business

**PC15.** identify sources for arranging money and associated financial and legal challenges

### *Customer Service*

To be competent, the user/individual on the job must be able to:

**PC16.** identify different types of customers

**PC17.** identify customer needs and address them appropriately

**PC18.** follow appropriate hygiene and grooming standards

### *Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC19.** create a basic biodata

**PC20.** search for suitable jobs and apply

**PC21.** identify and register apprenticeship opportunities as per requirement

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use basic spoken English language

**KU6.** Do and dont of effective communication

**KU7.** inclusivity and its importance

**KU8.** different types of disabilities and appropriate communication and behaviour towards PwD

**KU9.** different types of financial products and services





## Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> understand the significance of employability skills in meeting the job requirements                                                                                                                                                                | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                     | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC2.</b> identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                             | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC3.</b> explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc. | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                    | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC4.</b> speak with others using some basic English phrases or sentences                                                                                                                                                                                    | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC5.</b> follow good manners while communicating with others                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC6.</b> work with others in a team                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                                                                                                                                                               | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC7.</b> communicate and behave appropriately with all genders and PwD                                                                                                                                                                                      | -            | -               | -             | -          |
| <b>PC8.</b> report any issues related to sexual harassment                                                                                                                                                                                                     | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                                                                                                                                                            | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC9.</b> use various financial products and services safely and securely                                                                                                                                                                                    | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> calculate income, expenses, savings etc.                                             | -            | -               | -             | -          |
| <b>PC11.</b> approach the concerned authorities for any exploitation as per legal rights and laws | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                   | <b>4</b>     | <b>6</b>        | -             | -          |
| <b>PC12.</b> operate digital devices and use its features and applications securely and safely    | -            | -               | -             | -          |
| <b>PC13.</b> use internet and social media platforms securely and safely                          | -            | -               | -             | -          |
| <i>Entrepreneurship</i>                                                                           | <b>3</b>     | <b>5</b>        | -             | -          |
| <b>PC14.</b> identify and assess opportunities for potential business                             | -            | -               | -             | -          |
| <b>PC15.</b> identify sources for arranging money and associated financial and legal challenges   | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC16.</b> identify different types of customers                                                | -            | -               | -             | -          |
| <b>PC17.</b> identify customer needs and address them appropriately                               | -            | -               | -             | -          |
| <b>PC18.</b> follow appropriate hygiene and grooming standards                                    | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC19.</b> create a basic biodata                                                               | -            | -               | -             | -          |
| <b>PC20.</b> search for suitable jobs and apply                                                   | -            | -               | -             | -          |
| <b>PC21.</b> identify and register apprenticeship opportunities as per requirement                | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                  | <b>20</b>    | <b>30</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0101                   |
| <b>NOS Name</b>            | Employability Skills (30 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 2                               |
| <b>Credits</b>             | 1                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 22/01/2026                      |
| <b>Next Review Date</b>    | 22/01/2029                      |
| <b>NSQC Clearance Date</b> | 22/01/2026                      |

### Assessment Guidelines and Assessment Weightage

#### Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

#### Minimum Aggregate Passing % at QP Level : 70



## Qualification Pack

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

### Assessment Weightage

Compulsory NOS

| National Occupational Standards                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|---------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| ISC/N0008.Use basic health and safety practices at the workplace    | 30           | 50              | 0             | 20         | 100         | 10         |
| ISC/N0009.Work effectively with others                              | 30           | 50              | 0             | 20         | 100         | 10         |
| ISC/N1102.Carry out maintenance of measuring and control equipment  | 30           | 50              | -             | 20         | 100         | 35         |
| ISC/N1103.Perform inspection and calibration of measuring equipment | 30           | 50              | -             | 20         | 100         | 35         |
| DGT/VSQ/N0101.Employability Skills (30 Hours)                       | 20           | 30              | 0             | 0          | 50          | 10         |
| <b>Total</b>                                                        | <b>140</b>   | <b>230</b>      | <b>-</b>      | <b>80</b>  | <b>450</b>  | <b>100</b> |



## Qualification Pack

### Acronyms

|             |                                                             |
|-------------|-------------------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)                           |
| <b>NSQF</b> | National Skills Qualifications Framework                    |
| <b>QP</b>   | Qualifications Pack                                         |
| <b>TVET</b> | Technical and Vocational Education and Training             |
| <b>5S</b>   | Technique of maintaining orderliness “Japanese terminology. |
| <b>CP</b>   | Control Plan.                                               |
| <b>WI</b>   | Work Instructions.                                          |



## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |



## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |